1 MRAC for SISO systems with known parameters

Consider the SISO system

$$\begin{aligned} \dot{x} &= A\,x + B\,u \\ y &= C\,x \end{aligned}$$

with known matrices A, B, C and transfer function

$$G(s) = C\,(s\,I - A)^{-1}B = k\,\frac{N(s)}{D(s)}$$

where $N(s)$ is a monic polynomial of degree $\deg N(s) = m$ and $D(s)$ is a monic polynomial of degree $\deg D(s) = n \leq N$.

Also consider the reference model

$$\begin{aligned} \dot{x}_m &= A_m\,x_m + B_m\,r \\ y_m &= C_m\,x_m \end{aligned}$$

with known matrices A_m, B_m, C_m and transfer function

$$G_m(s) = C_m\,(s\,I - A_m)^{-1}B_m = k_m\,\frac{N_m(s)}{D_m(s)}$$

where $N_m(s)$ is a monic Hurwitz polynomial of degree $\deg N(s) = m_m$ and $D(s)$ is a monic Hurwitz polynomial of degree $\deg D(s) = n_m$.

Assume that the two systems have the same relative degree.

$$n - m = n_m - m_m$$

We will show that the feedback control law

$$u(s) = \theta_1^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} u(s) + \theta_2^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} y(s) + \theta_3 y(s) + c\,r(s)$$

where

- $\theta_1, \theta_2 \in \mathbb{R}^{N-1}$ and $\theta_3, c \in \mathbb{R}$
- $$\alpha_{N-2}(s) = \begin{array}{ll} [s^{N-2} \ldots s \quad 1] & \text{if } N-2 \geq 0 \\ 0 & \text{otherwise} \end{array}$$
- $\Lambda(s) = \Lambda_0(s)\,N_m(s)$ is a monic polynomial of degree $N-1$, given by

$$\Lambda(s) = s^{N-1} + \lambda_{N-2}s^{N-2} + \cdots + \lambda_1 s + \lambda_0$$

ensures that

$$\lim_{t\to\infty} y(t) - y_m(t) = 0$$

Proof.

First, derive the reference-to-output transfer function of the plant. Since

$$u(s) \;=\; \theta_1^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} u(s) + \theta_2^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} y(s) + \theta_3 y(s) + c\, r(s)$$

we obtain

$$\begin{aligned}
u(s) - \theta_1^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} u(s) &= \theta_2^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} y(s) + \theta_3 y(s) + c\, r(s) \\
\left(1 - \theta_1^T \frac{\alpha_{N-2}(s)}{\Lambda(s)}\right) u(s) &= \left(\theta_2^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} + \theta_3\right) y(s) + c\, r(s) \\
\frac{\Lambda(s) - \theta_1^T \alpha_{N-2}(s)}{\Lambda(s)} u(s) &= \frac{(\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) + c\, \Lambda(s)\, r(s)}{\Lambda(s)} \\
u(s) &= \frac{\cancel{\Lambda(s)}}{\Lambda(s) - \theta_1^T \alpha_{N-2}(s)} \frac{(\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) + c\, \Lambda(s)\, r(s)}{\cancel{\Lambda(s)}} \\
&= \frac{(\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) + c\, \Lambda(s)\, r(s)}{\Lambda(s) - \theta_1^T \alpha_{N-2}(s)}
\end{aligned}$$

It follows that

$$\begin{aligned}
y(s) &= k \frac{N(s)}{D(s)} u(s) \\
&= k \frac{N(s)}{D(s)} \left(\frac{(\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) + c\, \Lambda(s)\, r(s)}{\Lambda(s) - \theta_1^T \alpha_{N-2}(s)} \right) \\
D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) y(s) &= k\, N(s) [(\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) + c\, \Lambda(s)\, r(s)] \\
D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) y(s) &= k\, N(s) (\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) + c\, k\, N(s)\, \Lambda(s)\, r(s)
\end{aligned}$$

Moving $y(s)$ to the left-hand side gives

$$\begin{aligned}
D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) y(s) - k\, N(s) (\theta_2^T \alpha_{N-2}(s) + \theta_3) y(s) &= c\, k\, N(s)\, \Lambda(s)\, r(s) \\
[D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) - k\, N(s) (\theta_2^T \alpha_{N-2}(s) + \theta_3)] y(s) &= c\, k\, N(s)\, \Lambda(s)\, r(s)
\end{aligned}$$

The reference-to-output transfer function is therefore

$$\frac{y(s)}{r(s)} \;=\; \frac{c\, k\, N(s)\, \Lambda(s)}{D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) - k\, N(s) (\theta_2^T \alpha_{N-2}(s) + \theta_3)}$$

To ensure $\lim_{t \to \infty} y(t) - y_m(t) = 0$, the plant's closed-loop transfer function must match the reference model's transfer function.

$$k_m \frac{N_m(s)}{D_m(s)} \;=\; \frac{c\, k\, N(s)\, \Lambda(s)}{D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) - k\, N(s) (\theta_2^T \alpha_{N-2}(s) + \theta_3)}$$

Since all the polynomials are monic, we can set

$$k_m \;=\; c\, k$$

Also, recalling that $\Lambda(s) = \Lambda_0(s) N_m(s)$, we obtain

$$\frac{\cancel{N_m(s)}}{D_m(s)} \;=\; \frac{N(s)\, \Lambda_0(s) \cancel{N_m(s)}}{D(s)(\Lambda(s) - \theta_1^T \alpha_{N-2}(s)) - k\, N(s) (\theta_2^T \alpha_{N-2}(s) + \theta_3)}$$

It follows that

$$\begin{aligned}
D(s)(\Lambda(s)-\theta_1^T\alpha_{N-2}(s))-k\,N(s)(\theta_2^T\alpha_{N-2}(s)+\theta_3) &= D_m(s)N(s)\,\Lambda_0(s)\\
D(s)\Lambda(s)-D(s)\theta_1^T\alpha_{N-2}(s)-k\,N(s)\theta_2^T\alpha_{N-2}(s)-k\,N(s)\theta_3 &= D_m(s)N(s)\,\Lambda_0(s)\\
-D(s)\theta_1^T\alpha_{N-2}(s)-k\,N(s)\theta_2^T\alpha_{N-2}(s)-k\,N(s)\theta_3 &= -D(s)\Lambda(s)+D_m(s)N(s)\,\Lambda_0(s)\\
D(s)\theta_1^T\alpha_{N-2}(s)+k\,N(s)\theta_2^T\alpha_{N-2}(s)+k\,N(s)\theta_3 &= D(s)\Lambda(s)-D_m(s)N(s)\,\Lambda_0(s)
\end{aligned}$$

Since

$$\begin{aligned}
\deg(N(s)) &= m\\
\deg(D(s)) &= n\\
\deg(D_m(s)) &= n_m\\
\deg(\Lambda(s)) &= N-1\\
\deg(\Lambda_0(s)) &= \deg(\Lambda(s))-\deg(N_m(s))=N-1-m_m\\
\alpha_{N-2}(s) &= \begin{array}{ll}[s^{N-2}\ldots s \quad 1] & \text{if } N-2\geq 0\\ 0 & \text{otherwise}\end{array}
\end{aligned}$$

we can make the following observations.

- Since

$$\begin{aligned}
\deg(D(s)\theta_1^T\alpha_{N-2}(s)) &\leq n+N-2\\
\deg(k\,N(s)\theta_2^T\alpha_{N-2}(s)) &\leq m+N-2\\
\deg(k\,N(s)\theta_3) &\leq m
\end{aligned}$$

  and $m\leq n-1$, the left-hand side has degree $n+N-2$.

- Since

$$\begin{aligned}
\deg(D(s)\Lambda(s)) &= n+N-1\\
\deg(-D_m(s)N(s)\,\Lambda_0(s)) &= n_m+m+N-1-m_m\\
&= m+N-1+(n_m-m_m)\\
&= \not{n}+N-1+(n-\not{n})\\
&= N-1+n
\end{aligned}$$

  and both polynomials are monic, the leading term s^{n+N-1} cancels, so the left-hand side has degree $n+N-2$.

There are therefore $n+N-1$ equations to solve.

Since $\theta_1,\theta_2\in\mathbb{R}^{N-1}$ and $\theta_3\in\mathbb{R}$, there are $2N-2+1=2N-1$ unknowns. For the equations to have at least one solution, we require

$$\begin{aligned}
2N-1 &\geq n+N-1\\
N &\geq n
\end{aligned}$$

This inequality holds by assumption, and $N(s)$ and $D(s)$ are coprime. Therefore, at least one solution always exists.

To implement the control law

$$\begin{aligned} u(s) &= \theta_1^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} u(s) + \theta_2^T \frac{\alpha_{N-2}(s)}{\Lambda(s)} y(s) + \theta_3 y(s) + c\, r(s) \\ &= \begin{pmatrix} \theta_1^T & \theta_2^T & \theta_3 & c \end{pmatrix} \begin{pmatrix} \frac{\alpha_{N-2}(s)}{\Lambda(s)} u(s) \\ \frac{\alpha_{N-2}(s)}{\Lambda(s)} y(s) \\ y(s) \\ r(s) \end{pmatrix} \\ &= \theta^T \omega \end{aligned}$$

define

$$\begin{aligned} \omega_1 &= \frac{\alpha_{N-2}(s)}{\Lambda(s)} u(s) \\ \omega_2 &= \frac{\alpha_{N-2}(s)}{\Lambda(s)} y(s) \end{aligned}$$

The dynamics of ω_1 and ω_2 are therefore

$$\begin{aligned} \dot{\omega}_1 &= F\omega_1 + g\, u \\ \dot{\omega}_2 &= F\omega_2 + g\, y \end{aligned}$$

where

$$\begin{aligned} F &= \begin{pmatrix} -\lambda_{N-2} & -\lambda_{N-3} & \dots & -\lambda_1 & -\lambda_0 \\ 1 & 0 & & & \\ & & \ddots & & \\ & & & 1 & 0 \end{pmatrix} \\ g &= \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \end{aligned}$$

Indeed,

$$\begin{aligned} \omega_1(s) &= (s\, I - F)^{-1} g\, u(s) \\ &= \frac{1}{\det(s\, I - F)} (s\, I - F)^* \cdot e_1 \cdot u(s) \end{aligned}$$

where

$$\begin{aligned} (s\, I - F)^* &= \begin{pmatrix} s + \lambda_{N-2} & \lambda_{N-3} & \dots & \lambda_1 & \lambda_0 \\ -1 & s & & & \\ 0 & -1 & \ddots & & \\ & & \ddots & s & \\ & & & -1 & s \end{pmatrix}^* \\ &= \begin{pmatrix} s^{n-1} & * \\ -(-s^{n-2}) & * \\ \vdots & \vdots \\ 1 & * \end{pmatrix} \end{aligned}$$

and

$$\begin{aligned}\det(s\,I-F) &= (s+\lambda_{N-2})\,s^{n-1}-\lambda_{N-3}(-s^{n-2})+\cdots+\lambda_0\\ &= s^n+\lambda_{N-2}s^{n-1}+\lambda_{N-3}s^{n-2}+\cdots+\lambda_0\\ &= \Lambda(s)\end{aligned}$$

It follows that

$$\begin{aligned}\omega(s) &= \frac{1}{\det(s\,I-F)}(s\,I-F)^* e_1 u(s)\\ &= \frac{1}{\Lambda(s)}\begin{pmatrix} s^{n-1}\\ -(-s^{n-2})\\ \vdots\\ 1\end{pmatrix}u(s)\\ &= \frac{\alpha_{n-1}(s)}{\Lambda(s)}u(s)\end{aligned}$$

The same reasoning establishes the dynamics of ω_2.

The overall system is therefore

$$\begin{aligned}\dot{x} &= A\,x+B\,u\\ \dot{\omega}_1 &= F\omega_1+g\,u\\ \dot{\omega}_2 &= F\omega_2+g\,y\end{aligned}$$

Recalling that $u=\theta^T\omega$ and $y=C\,x$, we obtain

$$\begin{aligned}\dot{x} &= A\,x+B\,u\\ &= A\,x+B\,\theta^T\omega\\ \dot{\omega}_1 &= F\omega_1+g\,u\\ &= F\omega_1+g\,\theta^T\omega\\ \dot{\omega}_2 &= F\omega_2+g\,y\\ &= F\omega_2+g\,C\,x\end{aligned}$$

The system can therefore be written in matrix form as follows.

$$\begin{aligned}\begin{pmatrix}\dot{x}\\ \dot{\omega}_1\\ \dot{\omega}_2\end{pmatrix} &= \begin{pmatrix} A\,x+B\,\theta^T\omega\\ F\omega_1+g\,\theta^T\omega\\ F\omega_2+g\,C\,x\end{pmatrix}\\ &= \begin{pmatrix} A\,x\\ F\omega_1\\ F\omega_2+g\,C\,x\end{pmatrix}+\begin{pmatrix} B\,\theta^T\omega\\ g\,\theta^T\omega\\ 0\end{pmatrix}\\ &= \begin{pmatrix} A & 0 & 0\\ 0 & F & 0\\ g\,C & 0 & F\end{pmatrix}\begin{pmatrix} x\\ \omega_1\\ \omega_2\end{pmatrix}+\begin{pmatrix} B\\ g\\ 0\end{pmatrix}\theta^T\omega\\ &= A_{\rm cl}\begin{pmatrix} x\\ \omega_1\\ \omega_2\end{pmatrix}+B_{\rm cl}\theta^T\omega\end{aligned}$$

□

2 Assignment 5

Consider the second-order system

$$y(s) \;=\; \frac{s+2}{s^2-10\,s+5}u(s)$$

and the reference model

$$y_m(s) \;=\; \frac{1}{s+1}r(s)$$

Let $\Lambda(s)=s+2$.

Verify that all assumptions for the MRAC design are satisfied.

Simulate the system response assuming

$$r(t) \;=\; E_1\sin(\omega_1 t)+E_2\sin(\omega_2 t)$$

where $E_1, E_2 \neq 0$ and $\omega_1 \neq \omega_2 \neq 0$.

3 Plant and reference model dynamics

The Simulink Transfer Fcn block does not let us specify the initial output directly. We therefore derive the plant's state-space matrices A, B, C for the implementation.

Using observable canonical form, we obtain

$$\begin{aligned} A &= \begin{pmatrix} 10 & -5 \\ 1 & 0 \end{pmatrix} \\ B &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{aligned}$$

Setting

$$C = \begin{pmatrix} c_1 & c_2 \end{pmatrix}$$

gives

$$\begin{aligned} s\,I - A &= \begin{pmatrix} s-10 & 5 \\ -1 & s \end{pmatrix} \\ \det(s\,I - A) &= s\,(s-10) + 5 \\ &= s^2 - 10\,s + 5 \\ (s\,I - A)^* &= \begin{pmatrix} s-10 & 5 \\ -1 & s \end{pmatrix}^* \\ &= \begin{pmatrix} s & -5 \\ 1 & s-10 \end{pmatrix} \\ C\,(s\,I - A)^{-1} B &= \frac{C\,(s\,I - A)^* B}{\det(s\,I - A)} \\ &= \frac{1}{s^2 - 10\,s + 5} \begin{pmatrix} c_1 & c_2 \end{pmatrix} \begin{pmatrix} s & -5 \\ 1 & s-10 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{s^2 - 10\,s + 5} \begin{pmatrix} c_1 & c_2 \end{pmatrix} \begin{pmatrix} s \\ 1 \end{pmatrix} \\ &= \frac{c_1 s + c_2}{s^2 - 10\,s + 5} \end{aligned}$$

To ensure

$$C\,(s\,I - A)^{-1} B = \frac{s+2}{s^2 - 10\,s + 5}$$

we require

$$\begin{aligned} c_1 &= 1 \\ c_2 &= 2 \end{aligned}$$

In summary, the plant dynamics are

$$\begin{aligned} \dot{x} &= \begin{pmatrix} 10 & -5 \\ 1 & 0 \end{pmatrix} x + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u \\ y &= \begin{pmatrix} 1 & 2 \end{pmatrix} x \end{aligned}$$

4 Controller design

The denominator of $G_m(s)$ is Hurwitz, and $G(s)$ and $G_m(s)$ have the same relative degree. The assumptions for the MRAC design are therefore satisfied.

Since $\deg(D(G(s))) = 2$, set $N = 2$ and define $\theta_1, \theta_2 \in \mathbb{R}^{N-1} = \mathbb{R}$, $\theta_3 \in \mathbb{R}$, and $c \in \mathbb{R}$. Given

$$\begin{aligned} u(s) &= \theta_1 \frac{\alpha_0(s)}{\Lambda(s)} u(s) + \theta_2 \frac{\alpha_0(s)}{\Lambda(s)} y(s) + \theta_3 y(s) + c\, r(s) \\ &= \theta_1 \frac{1}{s+2} u(s) + \theta_2 \frac{1}{s+2} y(s) + \theta_3 y(s) + c\, r(s) \end{aligned}$$

we now derive the reference-to-output transfer function of the plant.

First, express $u(s)$ in terms of $y(s)$ and $r(s)$.

$$\begin{aligned} u(s) &= \theta_1 \frac{1}{s+2} u(s) + \theta_2 \frac{1}{s+2} y(s) + \theta_3 y(s) + c\, r(s) \\ u(s) - \theta_1 \frac{1}{s+2} u(s) &= \theta_2 \frac{1}{s+2} y(s) + \theta_3 y(s) + c\, r(s) \\ \left(1 - \theta_1 \frac{1}{s+2}\right) u(s) &= \theta_2 \frac{1}{s+2} y(s) + \theta_3 y(s) + c\, r(s) \\ \frac{s+2-\theta_1}{s+2} u(s) &= \frac{\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)}{s+2} \\ u(s) &= \frac{s \neq 2}{s+2-\theta_1} \frac{\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)}{s \neq 2} \\ &= \frac{\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)}{s+2-\theta_1} \end{aligned}$$

Substituting into the dynamics of $y(s)$ gives

$$\begin{aligned} y(s) &= \frac{s+2}{s^2 - 10\,s + 5} u(s) \\ &= \frac{s+2}{s^2 - 10\,s + 5} \frac{\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)}{s+2-\theta_1} \\ &= \frac{(s+2)[\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)]}{s^3 + 2s^2 - \theta_1 s^2 - 10 s^2 - 20 s + 10\theta_1 s + 5s + 10 - 5\theta_1} \\ &= \frac{(s+2)[\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)]}{s^3 + (2 - \theta_1 - 10) s^2 + (-20 + 10\theta_1 + 5) s + 10 - 5\theta_1} \\ &= \frac{(s+2)[\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)]}{s^3 + (-\theta_1 - 8) s^2 + (10\theta_1 - 15) s + 10 - 5\theta_1} \end{aligned}$$

Now isolate $y(s)$:.

$$\begin{aligned} [s^3 + (-\theta_1 - 8)s^2 + (10\theta_1 - 15)s + 10 - 5\theta_1] y(s) &= (s+2)[\theta_2 y(s) + (s+2)\theta_3 y(s) + c\,(s+2) r(s)] \\ [s^3 + (-\theta_1 - 8)s^2 + (10\theta_1 - 15)s + 10 - 5\theta_1] y(s) &= (s+2)\theta_2 y(s) + (s+2)^2 \theta_3 y(s) + c(s+2)^2 r(s) \\ [s^3 + (-\theta_1 - 8)s^2 + (10\theta_1 - 15)s + 10 - 5\theta_1] y(s) &= [(s+2)\theta_2 + (s^2 + 4s + 4)\theta_3] y(s) + c(s+2)^2 r(s) \\ [s^3 + (-\theta_1 - 8)s^2 + (10\theta_1 - 15)s + 10 - 5\theta_1] y(s) &= [\theta_2 s + 2\theta_2 + \theta_3 s^2 + 4\theta_3 s + 4\theta_3] y(s) + c(s+2)^2 r(s) \\ [s^3 + (-\theta_1 - 8)s^2 + (10\theta_1 - 15)s + 10 - 5\theta_1] y(s) &= [\theta_3 s^2 + (\theta_2 + 4\theta_3) s + 2\theta_2 + 4\theta_3] y(s) + c(s+2)^2 r(s) \end{aligned}$$

It follows that

$$[s^3 + (-\theta_1 - 8 - \theta_3)s^2 + (10\theta_1 - 15 - \theta_2 - 4\theta_3)s + 10 - 5\theta_1 - 2\theta_2 - 4\theta_3] y(s) = c(s+2)^2 r(s)$$

$$y(s) = \frac{c(s+2)^2}{s^3 + (-\theta_1 - 8 - \theta_3)s^2 + (10\theta_1 - 15 - \theta_2 - 4\theta_3)s + 10 - 5\theta_1 - 2\theta_2 - 4\theta_3} r(s)$$

To ensure $y(t) \to y_m(t)$, the two reference-to-output transfer functions must match.

$$\frac{c(s+2)^2}{s^3+(-\theta_1-8-\theta_3)s^2+(10\theta_1-15-\theta_2-4\theta_3)s+10-5\theta_1-2\theta_2-4\theta_3} \;=\; \frac{1}{s+1}$$

It follows that

$$c \;=\; 1$$

and

$$\begin{aligned}
(s+1)(s+2)^2 &= s^3+(-\theta_1-8-\theta_3)s^2+(10\theta_1-15-\theta_2-4\theta_3)s+10-5\theta_1-2\theta_2-4\theta_3\\
(s+1)(s^2+4s+4) &= s^3+(-\theta_1-8-\theta_3)s^2+(10\theta_1-15-\theta_2-4\theta_3)s+10-5\theta_1-2\theta_2-4\theta_3\\
s^3+4s^2+4s+s^2+4s+4 &= s^3+(-\theta_1-8-\theta_3)s^2+(10\theta_1-15-\theta_2-4\theta_3)s+10-5\theta_1-2\theta_2-4\theta_3\\
s^3+5s^2+8s+4 &= s^3+(-\theta_1-8-\theta_3)s^2+(10\theta_1-15-\theta_2-4\theta_3)s+10-5\theta_1-2\theta_2-4\theta_3
\end{aligned}$$

Therefore,

$$\begin{aligned}
5 &= -\theta_1-8-\theta_3\\
8 &= 10\theta_1-15-\theta_2-4\theta_3\\
4 &= 10-5\theta_1-2\theta_2-4\theta_3
\end{aligned}$$

$$\begin{aligned}
\theta_1+\theta_3 &= -8-5\\
-10\theta_1+\theta_2+4\theta_3 &= -15-8\\
5\theta_1+2\theta_2+4\theta_3 &= 10-4
\end{aligned}$$

$$\begin{aligned}
\theta_1+\theta_3 &= -13\\
-10\theta_1+\theta_2+4\theta_3 &= -23\\
5\theta_1+2\theta_2+4\theta_3 &= 6
\end{aligned}$$

$$\begin{aligned}
\theta_1+\theta_3 &= -13\\
10\theta_1-\theta_2-4\theta_3 &= 23\\
5\theta_1+2\theta_2+4\theta_3 &= 6
\end{aligned}$$

$$\begin{pmatrix} 1 & 0 & 1 & -13 \\ 10 & -1 & -4 & 23 \\ 5 & 2 & 4 & 6 \end{pmatrix}$$

$$e_{21}(-10)\downarrow e_{31}(-5)$$

$$\begin{pmatrix} 1 & 0 & 1 & -13 \\ 0 & -1 & -14 & 153 \\ 0 & 2 & -1 & 71 \end{pmatrix}$$

$$\downarrow e_{32}(2)$$

$$\begin{pmatrix} 1 & 0 & 1 & -13 \\ 0 & -1 & -14 & 153 \\ 0 & 0 & -29 & 377 \end{pmatrix}$$

$$\begin{aligned}
\theta_1+\theta_3 &= -13\\
-\theta_2-14\theta_3 &= 153
\end{aligned}$$

$$
\begin{aligned}
-29\theta_3 &= 377 \\
\\
\theta_1 &= -13 - \theta_3 \\
\theta_2 &= -153 - 14\theta_3 \\
\theta_3 &= -\frac{377}{29} \\
\\
\theta_1 &= -13 + 13 \\
\theta_2 &= -153 + 14 \cdot 13 \\
\theta_3 &= -13 \\
\\
\theta_1 &= 0 \\
\theta_2 &= 29 \\
\theta_3 &= -13
\end{aligned}
$$

The resulting control law is

$$
\begin{aligned}
u(s) &= \theta_1 \frac{1}{s+2} u(s) + \theta_2 \frac{1}{s+2} y(s) + \theta_3 y(s) + c\, r(s) \\
&= 29 \frac{1}{s+2} y(s) - 13 y(s) + r(s)
\end{aligned}
$$

Define

$$
\omega(s) = \frac{1}{s+2} y(s)
$$

Then

$$
\begin{aligned}
u(s) &= 29\,\omega(s) - 13\,y(s) + r(s) \\
u(t) &= 29\,\omega(t) - 13\,y(t) + r(t)
\end{aligned}
$$

where the dynamics of $\omega(t)$ are given by

$$
\begin{aligned}
s\,\omega_2(s) + 2\omega(s) &= y(s) \\
s\,\omega_2(s) &= -2\omega(s) + y(s) \\
\dot{\omega}(t) &= -2\omega(t) + y(t)
\end{aligned}
$$

5 Simulations

5.1 $E_1 = 1, \omega_1 = 1\,\frac{\mathbf{rad}}{s}, E_2 = 0.6, \omega_2 = 3\frac{\mathbf{rad}}{s}$

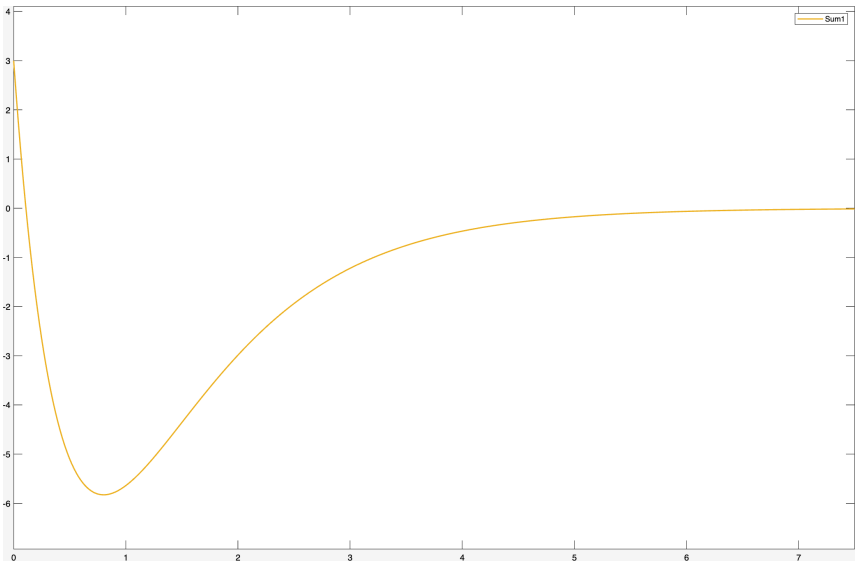
If

$$r(t) \;=\; \sin(t) + 0.6\sin(3t)$$

and

$$x(0) \;=\; \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

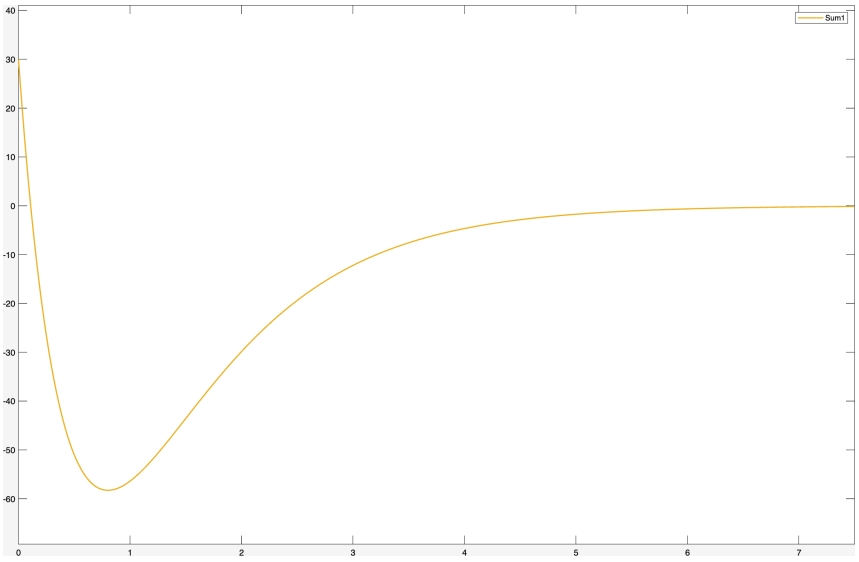
the error converges in approximately $6.8s$, reaching a minimum of -5.83.



With the alternative initial condition

$$x(0) \;=\; \begin{pmatrix} 10 \\ 10 \end{pmatrix}$$

the error converges in approximately $6.8s$, reaching a minimum of -58.3.



5.2 $E_1 = 2, \omega_1 = 0.5\,\frac{\mathbf{rad}}{s}, E_2 = 1, \omega_2 = 2\frac{\mathbf{rad}}{s}$

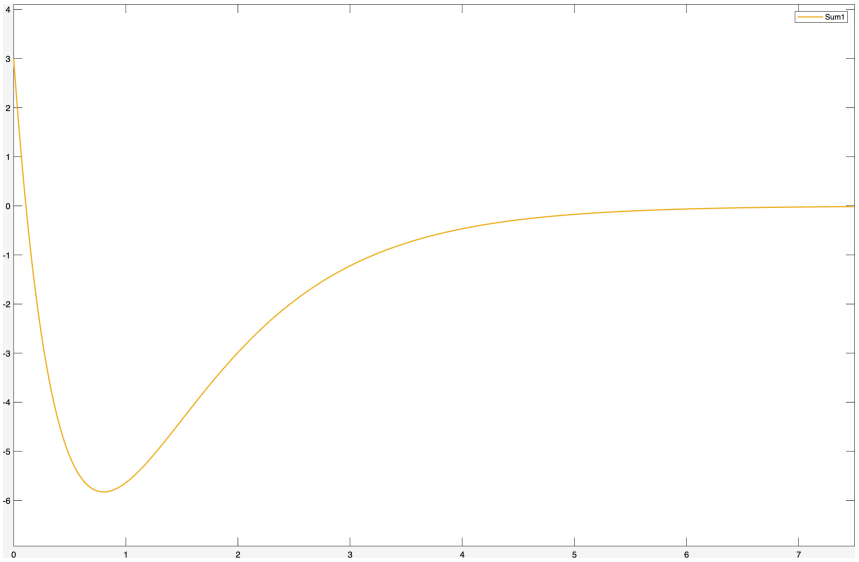
If

$$r(t) \;=\; 2\sin(0.5t) + \sin(2t)$$

and

$$x(0) \;=\; \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

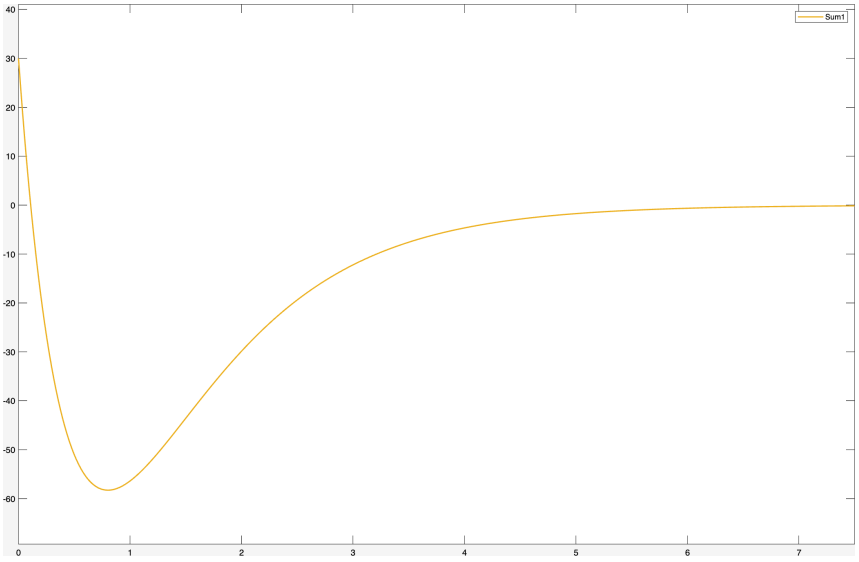
the error converges in approximately $6.8s$, reaching a minimum of -5.83.



With the alternative initial condition

$$x(0) \;=\; \begin{pmatrix} 10 \\ 10 \end{pmatrix}$$

the error converges in approximately $6.8s$, reaching a minimum of -58.3.



In these simulations, the convergence of the error to zero depends only on the initial state.